

R Command Reference Sheet

Clinical Data Analysis in R — Phase I | Neudata · #ClearDataClearImpact

A one-stop "cheat sheet" of the commands used across the five days. Keep it beside you during practicals.

1. Getting started

Task	Command
Assign a value to an object	<code>x <- 5</code>
Combine values into a vector	<code>c(1, 2, 3)</code>
Install a package (once)	<code>install.packages("tidyverse")</code>
Load a package (each session)	<code>library(tidyverse)</code>
Get help on a function	<code>?mean</code> or <code>help(mean)</code>
See current working directory	<code>getwd()</code>
List objects in memory	<code>ls()</code>

2. Importing data

Task	Command
Read a CSV	<code>read_csv("Data/file.csv")</code>
Read CSV, set missing codes	<code>read_csv("f.csv", na = c("", "NA", "999"))</code>
Read an Excel sheet	<code>read_excel("Data/file.xlsx", sheet = "data")</code>
Save an R object (keeps types)	<code>saveRDS(df, "Data/df.rds")</code>
Reload an R object	<code>df <- readRDS("Data/df.rds")</code>
Write a CSV	<code>write_csv(df, "out.csv")</code>

3. Inspecting data

Task	Command
Dimensions	<code>dim(df), nrow(df), ncol(df)</code>
Variable names	<code>names(df)</code>
Compact structure	<code>glimpse(df)</code>
First / last rows	<code>head(df), tail(df)</code>
Spreadsheet viewer	<code>View(df)</code>
One column as a vector	<code>df\$age</code>
Quick numeric summary	<code>summary(df\$age)</code>

4. Cleaning & wrangling (dplyr / stringr)

Task	Command
Keep rows meeting a condition	<code>filter(df, age >= 18)</code>
Keep/rename columns	<code>select(df, age, sex)</code>
Create / change a column	<code>mutate(df, bmi = weight_kg/(height_cm/100)^2)</code>
Remove duplicate rows	<code>distinct(df)</code>
Recode by rules	<code>case_when(x == "M" ~ "Male", TRUE ~ NA_character_)</code>
Trim whitespace	<code>str_trim(x)</code>
Lower case	<code>str_to_lower(x)</code>
Blank to NA	<code>na_if(x, "")</code>
Conditional value	<code>if_else(age > 110, NA_real_, age)</code>
Apply across columns	<code>mutate(across(where(is.character), str_trim))</code>
Bin a numeric var	<code>cut(bmi, breaks = c(-Inf, 18.5, 25, 30, Inf), labels = ...)</code>
Make a factor	<code>factor(sex, levels = c("Female", "Male"))</code>
Ordered factor	<code>factor(edu, levels = c(...), ordered = TRUE)</code>
Parse mixed dates	<code>lubridate::parse_date_time(x, c("ymd", "dmy"))</code>
The pipe	<code>`df</code>

5. Descriptive statistics

Task	Command
Mean / median (ignore NA)	<code>mean(x, na.rm = TRUE), median(x, na.rm = TRUE)</code>
SD / IQR	<code>sd(x, na.rm = TRUE), IQR(x, na.rm = TRUE)</code>
Quantiles	<code>quantile(x, na.rm = TRUE)</code>
Frequency table	<code>table(df\$sex)</code>
Proportions	<code>prop.table(table(df\$sex))</code>
Cross-tabulation	<code>table(df\$treatment_uptake, df\$diabetes)</code>
Row/column percentages	<code>prop.table(tab, margin = 1) (1 = rows, 2 = cols)</code>
Grouped summary	<code>`df</code>
Count by group	<code>count(df, education)</code>
Publication "Table 1"	<code>gtsummary::tbl_summary(df, by = treatment_uptake)</code>

6. Figures (ggplot2)

Plot	Command skeleton
Histogram	<code>ggplot(df, aes(age)) + geom_histogram()</code>
Bar chart	<code>ggplot(df, aes(education)) + geom_bar()</code>
Boxplot	<code>ggplot(df, aes(treatment_uptake, sbp_mmhg)) + geom_boxplot()</code>
Scatter + smoother	<code>ggplot(df, aes(bmi, sbp_mmhg)) + geom_point() + geom_smooth()</code>
Add labels	<code>+ labs(title = "...", x = "...", y = "...")</code>
Clean theme	<code>+ theme_minimal()</code>
Save at 300 dpi	<code>ggsave("Resources/fig.png", width = 7, height = 5, dpi = 300)</code>

7. Statistical tests

Test	Command
Normality (visual)	<code>qqnorm(x); qqline(x)</code>
Normality (test)	<code>shapiro.test(x)</code>
Two-group means	<code>t.test(age ~ treatment_uptake, data = df)</code>
Two-group (non-parametric)	<code>wilcox.test(age ~ treatment_uptake, data = df)</code>
>2 groups	<code>`aov(sbp_mmhg ~ bp_category, data = df)</code>
Post-hoc	<code>TukeyHSD(aov(...))</code>
Categorical association	<code>chisq.test(table(df\$a, df\$b))</code>
Small expected counts	<code>fisher.test(table(df\$a, df\$b))</code>
Correlation	<code>cor.test(df\$bmi, df\$sbp_mmhg) (add method = "spearman")</code>

8. Regression (logistic)

Task	Command
Fit logistic model	<code>glm(treatment_uptake ~ age + diabetes, data = df, family = binomial)</code>
Model summary	<code>summary(model)</code>
Odds ratios + 95% CI	<code>exp(cbind(OR = coef(model), confint(model)))</code>
Tidy OR table	<code>broom::tidy(model, exponentiate = TRUE, conf.int = TRUE)</code>
Compare models (LRT)	<code>anova(m1, m2, test = "LRT")</code>
Multicollinearity	<code>car::vif(model)</code>
Discrimination (AUC)	<code>pROC::roc(df\$y, fitted(model))</code>
Publication table	<code>gtsummary::tbl_regression(model, exponentiate = TRUE)</code>
Reproducibility footer	<code>sessionInfo()</code>

Tip: almost every summary function has an `na.rm = TRUE` argument. If a mean comes back as NA, missing values are the usual cause.